

Supplementary Information

For “Perovskite Photovoltaics in June 2026: Defect passivation, composition and phase control”

Havid Aqoma, October 9, 2026

This Supplementary Information contains the corpus construction and selection method (Note S1), the extraction and evidence-chain method (Note S2), the full reporting audit (Note S3), the corpus statistics (Note S4), the complete certified-record table (Note S5), the limitations (Note S6), and the automated gate report for this build (Note S7).

Note S1. Corpus construction and selection

The corpus came from an OpenAlex title-and-abstract search for perovskite AND (“solar cell” OR photovoltaic), restricted to June 2026 publication dates and to the work types article, preprint and review. That set was unioned with records from Semantic Scholar, Crossref and arXiv and de-duplicated by DOI; exclusion terms are applied in Python after retrieval, never inside the query. Of 623 works meeting the scope gate, 426 had a usable abstract and 167 met the depth-review threshold.

The depth tier is built on abstracts by decision, not by omission: a full-text retrieval probe succeeded for about 5% of a month’s papers, and the failure is publisher refusal of automated retrieval rather than indexing delay. Selection is a constrained draw over venue citation percentile and mechanism axis with per-axis minima and venue and institution share caps, seeded for reproducibility. Across 200 sensitivity draws the median Jaccard overlap of the selected set was 0.994, with 157 papers appearing in every draw.

Table S1. Records retrieved by source.

Source	Records retrieved
OpenAlex (keyed, cursor-paginated)	644
Semantic Scholar (bulk)	0
Crossref	200
arXiv	11
Corpus after de-duplication and topical post-filter	623
With a usable abstract	426
Non-English (machine translated)	13
Preprints	33

Table S2. Selection funnel.

Stage	Works
Passed the scope gate	623
Usable abstract (eligibility)	418
Depth tier (selected for close reading)	167
Extraction records surviving all guards	167
Dropped: no verifiable claim survived	0
Cited in the main text	75
Audited but not cited individually	92

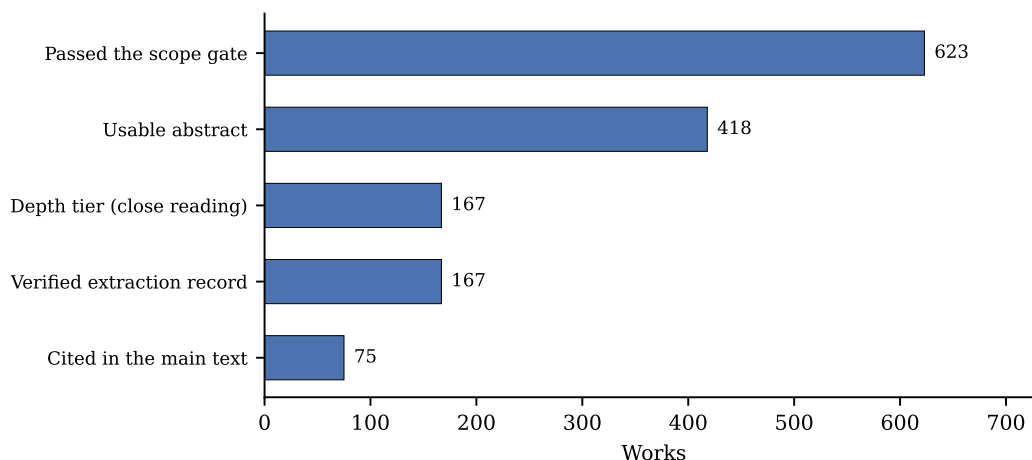


Figure S1. Corpus construction funnel.

Note S2. Extraction and the evidence chain

Every closely read paper was passed through a structured extraction step that returned numeric fields, each bound to a verbatim quotation from that paper’s own abstract. Three deterministic guards were then applied by script, never by the extracting model: first, the quotation must appear word for word in the source abstract. Second, the numeric value must appear inside its own quotation, and third, the quotation is truncated to 25 words before that check rather than after it, so a value can never be separated from the words that prove it. Any field failing a guard is nulled and counted, and a paper whose extraction left no verifiable claim was dropped from the cited set rather than shipped with an unprovable record (0 of 167 this month).

Independent re-verification of the shipped records, run as a separate script against the source abstracts, found every quotation present verbatim, none exceeding 25 words, and no numeric value missing from its own quotation. The abstract of the main text carries an additional gate: every numeral in it must exist either in the statistics file or in an extraction record, and each named scientific highlight is bound to one explicit record by its DOI, so a highlight cannot silently attach a real number to the wrong paper.

Note S3. The full reporting audit

This note reports how completely the June 2026 papers describe their own measurements. The audit reads abstracts, so it measures what authors chose to summarise, not what appears in an experimental section or in supporting information. Two tiers are covered: a corpus tier of 426 abstracts and a depth tier of 167 abstracts.

Because the hand-labelled validation set for the detector is not yet complete, every percentage here is a lower bound and is written as “detected in at least”, with no precision or recall figure attached. In the corpus tier an efficiency value was detected in at least 48.1% of abstracts, a stabilised or maximum-power-point-tracked efficiency in at least 10.1%, independent certification in at least 5.9%, an active or aperture area in at least 8.5%, and a complete open-circuit voltage, short-circuit current density and fill factor triplet in at least 3.1%. An ISOS stability-protocol label was detected in at least 1.6% and an explicit hysteresis statement in at least 2.6%. Table S3 gives both tiers side by side.

Table S3. Reporting-completeness detection rates. All values are lower bounds obtained by pattern detection on abstracts, not by human reading.

Reporting item	Corpus (n=426)	Depth (n=167)
Efficiency value stated	48.1%	59.9%
Stabilised or MPPT value	10.1%	16.8%
Independent certification	5.9%	8.4%

Reporting item	Corpus (n=426)	Depth (n=167)
Device area stated	8.5%	12.6%
Hysteresis or scan direction	2.6%	4.8%
ISOS protocol label	1.6%	3.0%
Complete Voc / Jsc / FF triplet	3.1%	0.6%

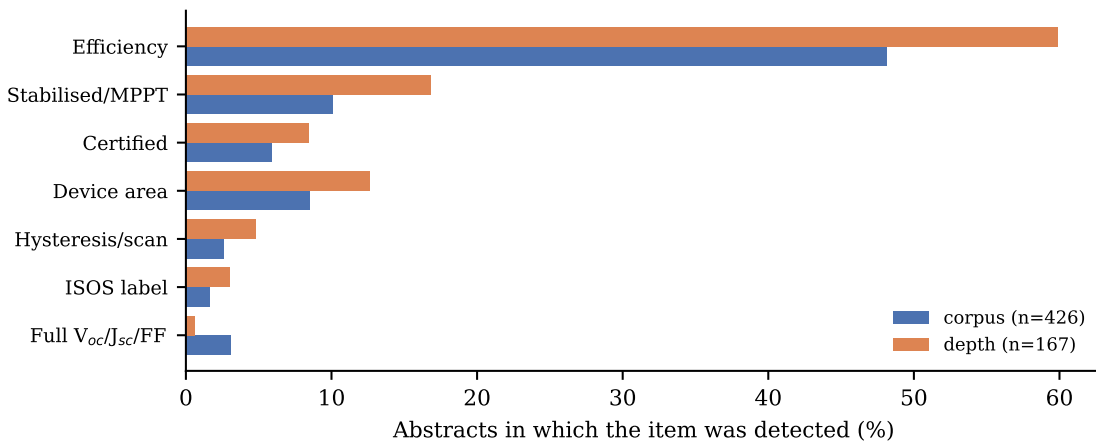


Figure S2. Reporting-completeness detection rates, corpus and depth tiers.

Two properties of these rates govern how they should be read. They are measured on abstracts, so they describe what authors chose to summarise rather than what was measured in the laboratory: an abstract omitting a device area is not evidence that the area went unrecorded. They are also lower bounds, because the hand-labelled validation set that would attach a precision and recall figure to each pattern is not yet complete. No percentage becomes exact until its precision reaches 0.85 against human labels, and until then every figure in this note carries that caveat without restating it.

Note S4. Corpus statistics

Table S4. Mechanism-axis distribution.

Mechanism axis	Corpus	Depth	Share of corpus
composition	268	41	43.0%
defects	113	46	18.1%
interfaces	107	30	17.2%
architecture	61	16	9.8%
stability	50	22	8.0%
scale up	24	12	3.9%

Axis assignment uses weighted keyword scoring over title and abstract across six fixed axes, with the title weighted twice, and mechanism centrality is the winning axis score normalised by that axis maximum. Selection into the depth tier scores venue citation percentile (0.40), mechanism centrality (0.35) and novelty (0.25), subject to a floor of 12 papers per axis, a preprint reservation, and per-venue and per-institution caps.

One caveat on the selection statistics is worth stating plainly: in this first issue the novelty term is constant, because it is defined against a cumulative history of extracted records that does not yet exist, so ranking reduces to venue percentile plus mechanism centrality. Repeating the selection under 200 perturbed weightings therefore returns a selection overlap of 0.994 with 157 papers selected in almost every draw. That number

should be read as a consequence of the flat novelty term, not as evidence that the selector is robust; it becomes informative from the third issue onward.

Note S5. Every certified efficiency reported in the month

Of 167 closely read papers, 14 reported an independently certified efficiency; the complete list follows, highest first, and each DOI is a live link.

Certified PCE (%)	Venue	Architecture	DOI
33.6	Science	tandem_2T	10.1126/science.aef5355
32.95	Nature Communications	tandem_2T	10.1038/s41467-026-73276-w
28.79	Angewandte Chemie International Edition	tandem_2T	10.1002/anie.4918900
27.28	Science Advances	unknown	10.1126/sciadv.aef6596
26.84	Nature Communications	unknown	10.1038/s41467-026-74107-8
26.8	Advanced Materials	p-i-n	10.1002/adma.73855
26.61	Advanced Energy Materials	module	10.1002/aenm.71159
26.44	ACS Energy Letters	p-i-n	10.1021/acsenergylett.6c01257
26.24	Advanced Materials	p-i-n	10.1002/adma.202522825
26.2	Advanced Materials	p-i-n	10.1002/adma.73656
26.12	Advanced Functional Materials	p-i-n	10.1002/adfm.76756
26.1	Nature Communications	unknown	10.1038/s41467-026-72581-8
25.56	Advanced Materials	unknown	10.1002/adma.73679
21.34	Advanced Energy Materials	p-i-n	10.1002/aenm.71216

Note S6. Limitations

Venue-impact-prioritised selection under-samples preprints, regional journals and non-English venues, so the depth subset is biased toward well-indexed English-language publishing. Work circulating chiefly on preprint servers or in regional venues can carry mechanism results and reporting habits this review never sees. Of the corpus, 33 records are preprints and 13 were machine translated from another language; translation can blur technical wording.

Because indexing lags publication, this review describes the indexed record of June 2026 as retrieved at build time, not the month itself. The lag is uneven, and work published late in the month is systematically under-represented relative to work published early. A later rebuild of the same month would retrieve a larger corpus, which is why no publication count appears in the title.

Four limitations are specific to this build: first, the reporting audit is unvalidated pending the human label set (Note S3). Second, the depth tier is built on abstracts by decision: a full-text retrieval probe succeeded for only 5% of the month’s papers, and a diagnostic across three months of differing age showed the failure is publisher refusal of automated retrieval rather than indexing delay, since a fourteen-month-old month retrieved worse than a two-month-old one. The audit therefore measures reported summaries, and the extraction records quote abstracts rather than full texts. Third, operational stability evidence is thin in an absolute sense: 6 papers reported a T80 lifetime and 5 named an ISOS protocol, which limits what any review can conclude about degradation this month. Fourth, this issue covers a single month, so nothing in it is a trend; the first month-over-month comparison becomes possible with the next issue.

Note S7. Automated gate report for this build

Gate	Status	Summary
G4	pass	em-dash 0, banned 0, self-report leaks 0
G1	pass	75 citations resolved, 0 unresolved
G2c-cite-count	pass	75 cited, target [60, 85], 40.3 words per citation
G2	pass	3020 words, 8 sections within 135% of ceiling
G2b-cite-order	pass	{“first_appearance_sequence”: [], “monotonic”: true}
G6	pass	{“hits”: []}
G3-abstract	pass	{“unverified”: [], “words”: 287, “placeholders_resolved”:
		[“A_MAX”, “H
G3c-abstract-physics	pass	{“implausible”: [], “sj_limit_pct”: 29.4}
G3d-illumination	pass	{“non_one_sun_values”: []}
G3b-abstract-form	pass	{“words”: 287, “band”: [260, 340], “citation_markers”: 0}
G7-abbrev	pass	{“expanded_automatically”:
		[“ISOS”, “PCE”, “SAM”, “PCE”, “MPPT”, “HTL”
G8-novelty	skip	no prior issue on record
G3e-body-numbers	pass	{“bound”: 130, “in_cited_abstract”: 14, “para_bound”: 0, “para_in_cite
G9a-notation	pass	{“residual_defects”: {}}, “ambiguous_for_human_review”:
		{}, “substituti
G9b-cite-links	pass	{“hyperlinked_markers”: 113, “unlinked_markers”: [], “works_without_do
G9c-figure-unique	pass	{“attached”:
		[“F1_certified_frontier.pdf”, “F4_stability_evidence.pdf”
G9d-model-names	pass	{“bare_slugs_found”: [], “declared”: [“qwen3.8-flash”, “gemini-3.8-fla
G10-title-block	pass	{“gaps_pt”: {“author_to_affil”: 14.42, “affil_to_date”: 22.52, “date_t
G5	pass	15 pages, 9.6 content

Gate definitions. G1 requires every citation marker in the main text to resolve to an extraction record carrying its own source quotation. G2 enforces length and structure ceilings and records a SHA-256 fingerprint of each drafted section, so a gate result is permanently bound to the text it judged. G4 enforces prose hygiene: em-dash count, a banned vocabulary list, and detection of leaked model self-commentary. G6 forbids first-person experimental claims, since this is a review of other groups’ work. The main-text abstract carries the additional numeric gate described in Note S2.